

Indian Institute of Technology Kharagpur

Mechanical Engineering Department

Autumn 2024-25 End Sem. Examination

Subject: MF41601: Soft Computing

Marks: 6+12+12+20=50; Time: 3 hours

Instructions: Answer all the questions. Assume suitable data, if necessary

Q. 1. Let us consider a constrained optimization problem of two variables: x_1 and x_2 as given below, where $0.0 \leq x_1 \leq 2.0$ and $2.0 \leq x_2 \leq 10.0$.

$$\text{Minimize } y=f(x_1, x_2)=x_1-x_2+x_1^2+x_2^2$$

subject to

$$x_1+2x_2 < 5.0,$$

$$11x_1^2-x_2^2 > 1.0$$

A Binary-Coded Genetic Algorithm (BCGA) is to be used to solve this optimization problem involving two real variables x_1 and x_2 , whose precision levels are to be set at 0.008 and 0.016, respectively. Design a suitable GA-string. The above constrained optimization problem is to be solved using the GA along with the concept of dynamic penalty. Assume the constants: $C=5.0$, $\alpha=2$, $\beta=3$ for the dynamic penalty function approach. Show one set of calculations (assuming generation as 1) for the values of the variables: $x_1=1.0$, $x_2=3.0$, to determine the value of dynamic penalty.

Q. 2. Fig. 1 shows a multi-layer feed-forward neural network to be used to model input-output relationships of a process having two inputs and one output. The neurons lying on the input, hidden and output layers have the transfer functions represented by $y = x$, $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ and $y = \frac{1}{1 + e^{-x}}$, respectively. The connecting weights between the input and hidden layers are represented by [V] and those between the hidden and output layers are denoted by [W]. The initial values of the connecting weights are assumed to be as follows: $v_{11}=0.1$, $v_{12}=0.2$, $v_{21}=0.3$, $v_{22}=0.2$, $w_{11}=0.3$, $w_{21}=0.4$.

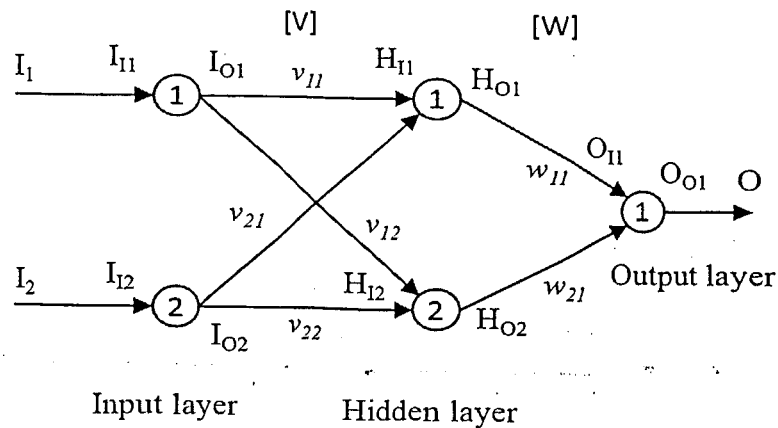


Fig. 1: A multi-layer feed-forward neural network.

Calculate the changes in [V] and [W] values using the Delta Rule of Back-Propagation algorithm. Take the learning rate $\eta=0.2$. Let us assume, one of the training scenarios as follows:

$I_1=0.4$, $I_2=0.6$, and output $O=0.7$. Show only one iteration. Note: Bias is set as zero at all the neurons.

Q. 3. A Neuro-Fuzzy System based on Takagi and Sugeno's approach, that is, adaptive neuro-fuzzy inference system (ANFIS) is to be developed for modelling a process with two inputs: I_1 and I_2 and one output: O . Two linguistic terms, such as N (Near) and F (Far) are utilized to represent first input I_1 . Similarly, the second input I_2 is to be expressed using two other linguistic terms, namely S (Small) and L (Large). For the first input I_1 , the membership function distributions for N and F are assumed to be of the forms: $y = \frac{1}{1+e^{-d_1x}}$ and $y = \frac{e^{-d_1x}}{1+e^{-d_1x}}$, respectively. Similarly, for the second input I_2 , the membership function distributions for S and L are assumed to be of the forms: respectively, $y = \frac{e^{d_2x}-e^{-d_2x}}{e^{d_2x}+e^{-d_2x}}$ and $y = \frac{2e^{-d_2x}}{e^{d_2x}+e^{-d_2x}}$, respectively. As there are two inputs and two linguistic terms have been used to represent each variable, there is a maximum of $2 \times 2 = 4$ possible rules. According to the first-order Takagi and Sugeno's model of fuzzy reasoning tool, the rules can be expressed as follows: $y^i = a_i I_1 + b_i I_2 + c_i$, where $i=1,2,3,4$; a_i, b_i, c_i are the coefficients of the rules. For the first, second, third and fourth rules, the values of a_i, b_i, c_i are assumed to be equal to (0.4,0.5,0.6), (0.5,0.6,0.7), (0.6,0.7,0.8) and (0.7,0.8,0.9), respectively. For simplicity, it has been assumed that the connecting weights $v_{11} = v_{12}$ and $v_{23} = v_{24}$. Moreover, d_1 and d_2 indicate the real values corresponding to the normalized weights $v_{11} = v_{12}$ and $v_{23} = v_{24}$, respectively. The values of d_1 and d_2 vary in the ranges: $1.5 \leq d_1 \leq 2.5$; $4.5 \leq d_2 \leq 5.5$. Assume the normalized weight values as follows: $[v_{11} = v_{12}, v_{23} = v_{24}]^T = [0.5, 0.6]$.

(i) Draw a schematic view of the said ANFIS.

(ii) Determine the output for the inputs: $I_1 = 2.0, I_2 = 6.5$.

Q. 4. Answer all the questions. Each question will carry 3 marks. Answer each question within the words limit of 100.

(a) Why does a Genetic Algorithm (GA) have a powerful global search but a relatively poor local search (e.g., as compared to steepest descent)? Explain.

(b) When is a Neural Network used, and in which problems is fuzzy logic controller used?

(c) How does a Neural Network train itself through optimization? Explain.

(d) Is Neuro-Fuzzy System basically a Fuzzy Logic Controller? Explain.

(e) Soft Computing is basically the combined techniques (e.g, Neuro-Fuzzy System, Genetic-Fuzzy System, Genetic-Neural System) only. Justify the statement.
